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Cultured *Cordyceps sinensis* polysaccharides attenuate cyclophosphamide-induced intestinal barrier injury in mice

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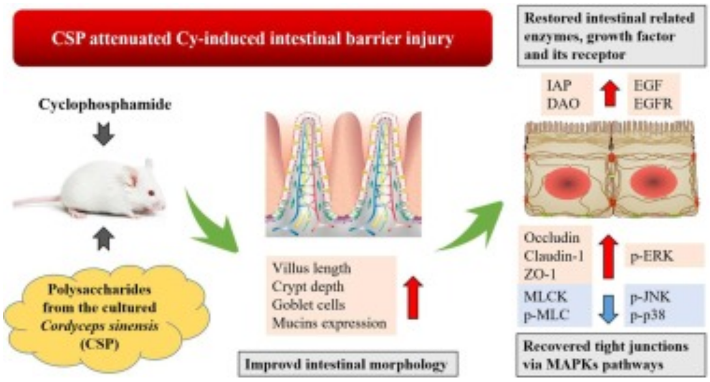
Highlights

- Cultured Cordyceps sinensis polysaccharides (CSP) improved intestinal morphology.
- CSP restored intestinal related enzymes, growth factor and its receptor.
- CSP upregulated tight junction proteins expression and downregulated MLCK, p-MLC.
- CSP recovered tight junctions via MAPKs pathways.
- CSP could attenuate cyclophosphamide-induced intestinal barrier injury in mice.

Abstract

Intestine is not only the largest digestive and absorbing organ, but also an important defensive barrier of the body. Maintaining healthy intestinal barrier is a prerequisite for intestinal function. In this study, polysaccharides from the cultured Cordyceps sinensis (CSP) were orally given to cyclophosphamide (Cy)-treated mice to investigate its protective effect on intestinal barrier injury. Results showed that CSP had positive effects on intestinal morphology, including restoring villus length and crypt depth as well as improving quantity of goblet cells and mucins expression. Furthermore, CSP enhanced intestinal alkaline phosphatase (IAP), diamine oxidase (DAO), epidermal growth factor (EGF) and its receptor (EGFR). In addition, occludin, claudin-1 and zonula occludins (ZO)-1 were upregulated and myosin light chain kinase (MLCK), p-MLC were downregulated by CSP. Moreover, CSP promoted p-ERK and inhibited p-JNK, p-p38. These findings suggested that CSP could attenuate Cy-induced intestinal barrier injury which maybe associated with mitogen-activated protein kinases (MAPKs) pathways.

Graphical abstract



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Introduction

Intestinal epithelium represents the single largest interface between our bodies and the external environment (Menard, Cerf-Bensussan, & Heyman, 2010). It is not only the largest digestive and absorbing organ, but also an important defensive barrier of the body (Walker, Thompson, & Battle, 2014). Most intestinal mucosal surfaces are covered with protective mucus which are mainly synthesized and secreted by intestinal goblet cells (Kim & Khan, 2013). There are also various enzymes and growth factors distributed in the intestine that play important roles in maintaining the integrity of intestinal barrier, including intestinal alkaline phosphatase (IAP), diamine oxidase (DAO), epidermal growth factor (EGF) and its receptor (EGFR). IAP is expressed on apical surfaces of microvillus membranes of enterocytes (Fawley & Gourlay, 2016) and DAO is widely present in the intestinal mucosa cells (Yu et al., 2018). They can be used to evaluate the integrity of intestinal barrier. EGF, a gastrointestinal mucosal protective factor, exerts various biological functions by binding to its receptor EGFR. It plays an important role in promoting cell growth and differentiation (Herbst, 2004). In addition to its growth promoting effect, it can effectively protect the intestinal epithelial barrier function from harmful stimuli, such as oxidative stress, ethanol and acetaldehyde (Seo et al., 2014).

Tight junctions are component of the apical junctional complex that seal the paracellular space between epithelial cells where claudins, zonula occludins (ZO), occludin and F-actin interact (Edens & Parkos, 2000). Myosin light chain kinase (MLCK) is associated with the perijunctional actomyosin ring, whose activation will increase tight junction permeability (Turner, 2009). MLCK-mediated phosphorylation of MLC would lead to the contraction of actin-myosin filaments, alteration of tight junction protein localization and expression as well as functional openings of tight junction barrier (Cunningham & Turner, 2012). Mitogen-activated protein kinases (MAPKs) are Ser/Thr protein kinases which include three classic subfamilies in mammalian cells: the extracellular signal-regulated kinases (ERKs), c-Jun N-terminal kinases (JNKs), and p38 MAPKs. These signaling pathways control fundamental cellular responses, such as cell proliferation, differentiation and apoptosis (Xiang et al., 2017). They are able to regulate tight junction paracellular transport through up- or downregulating several tight junction proteins expression and thus altering the molecular composition within tight junction complexes (Gonzalez-Mariscal, Tapia, & Chamorro, 2008). Preserving the integrity of intestinal barrier represents a critical task.

Cyclophosphamide (Cy), an oxazaphosphorine-substituted nitrogen mustard alkylating agent, has powerful cytotoxic and immunosuppressive effects (Ahlmann & Hempel, 2016). It is widely used to treat malignancies and severe manifestations of autoimmune diseases in clinical practice and trials (Emadi, Jones, & Brodsky, 2009). Besides, the well-known toxicity profile of Cy, resembling that of other chemotherapeutic agents, includes increased risk of infections, acute gastrointestinal mucosal damage and so on (Duncan & Grant, 2003). Therefore, Cy is always used to establish immunosuppressed and intestinal injury animal models (Shi et al., 2017, Xiang et al., 2018, Xie et al., 2015, Yu et al., 2014, Yu et al., 2015).

Cordyceps sinensis (*C. sinensis*) has been commonly known as a valued Chinese traditional herb for thousands of years. Polysaccharides are considered to be one of the major bioactive constituents in *C. sinensis* (Wang et al., 2012) and used as the ingredient of functional foods. The latest study of our lab revealed polysaccharides from natural *C. sinensis* were mainly composed of glucose, with α-1,4-glucose as the backbone (Wang et al., 2017). In addition, natural *C. sinensis* polysaccharides were demonstrated to have the ability of enhancing intestinal immunity in immunosuppressed mice (Chen, Wang, Fang, Dong, & Nie, 2019). However, due to the limited productivity and high cost of natural *C. sinensis*, several *Cordyceps* fungi were isolated from wild *C. sinensis* and fermented to produce cultured *C. sinensis* polysaccharides. Recently, we purified one kind of polysaccharides from the cultured *C. sinensis* (CSP), with a molecular weight of 28 kDa approximately. CSP were mainly comprising glucose, galactose, and mannose, with 1,4-glucose and 1,4-galactose as the main chain, which were different from natural *C. sinensis* polysaccharides (Wang et al., 2017). It has been well established that structure is the foundation of function, and different structures determine diverse bioactivities. At present, whether CSP have a positive function on the intestine remains unclear. Therefore, in this study, we focused on the intestinal mucosa to explore the protective effect of CSP on Cy-induced intestinal barrier injury.

Section snippets

Materials and reagents

Cultured *Cordyceps sinensis* was purchased from Guoyao Company (Jiangxi, China). Cyclophosphamide was purchased from Sigma-Aldrich (St. Louis, MO, USA). Levamisole hydrochloride (LH) was from Shandong Renhe Church Pharmaceutical Co., Ltd. (Tancheng, Shandong, China). Alkaline Phosphatase Assay Kit was purchased from Beyotime Biotechnolog (Shanghai, China). Micro Diamine Oxidase Assay Kit was purchased from Beijing Solarbio Science & Technology Co., Ltd (Beijing, China). Mouse EGF, EGFR ELISA …

Effect of CSP on body weight

Animal body weight was monitored every day half an hour before the experiment. As shown in Fig. 2, all groups except NC group showed a dramatic decrease in body weight 24 h after the last intraperitoneal injection of Cy (Day 4, $p < 0.01$). The weight of three CSP treated groups mice began to restore in Day 6 and kept rising during the remaining experimental time, so was the mice in PC group. However, unlike this, the weight loss of mice in M group lasted until Day 8 and started to recover which …

Discussion

In an adult human being, the gut epithelium covers a surface area of perhaps 300 m^2 when villi, microvilli, crypts and folds are taken into account, thus intestinal barrier function is a key component in the arsenal of defense mechanisms required to prevent infection and inflammation (Brandtzaeg, 2011, Peterson and Artis, 2014). The purpose of present study was to examine the protective effect of CSP on intestinal barrier injury induced by Cy.

During the experiment, Cy-induced mice exhibited …

Ethics statements

All animals used in this experiment were approved by the institutional animal care and use committee, Nanchang University. Additionally, these mice were cared for in accordance with the Guidelines for the Care and Use of Laboratory Animals published by the U.S. National Institutes of Health (NIH Publication 85-23, 1996). …

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Declaration of Competing Interest

The authors declare no competing financial interest. …

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